



What is Docker?

The benefit of a container is that it provides, from top to bottom, an identical environment in which you can run your Python script. You can install required software and packages in the container. This provide a silo in which the running code is kept separate from the host machine. Therefore, the underlying OS has no effect on a container or the task it performs. Distributing your Python code on a container guarantees a more similar runtime environment than a virtual environment made with traditional Python tools.

But before we can begin creating a Docker container, we first need a Python script to run in it. Over the next few pages, we will use Flask to build a simple website. We will then deploy this site in a Docker container on the internet via a cloud provider.

Reading Question

Select the **best** definition of a container.

- ☒ A container a separate environment when programs can be run independent from the underlying operating system.
- ☐ A container is another name for a virtual environment, where Python packages are kept separate from the system packages.
- ☐ A container exists when a host machine creates a virtual computer so programs can be run separate from the underlying operating system.
- ☐ A container is a partition on a hard drive where another operating system can be installed.

The correct answer is:

A container a separate environment when programs can be run independent from the underlying operating system.

- A container is not a virtualized computer. This is called a virtual machine. Containers only need the resources to perform a particular task. Therefore, they do not require an entire virtual computer. Because of this, containers are more performant than a virtual machine.
- Virtual environments are a development tool to keep Python packages for a project separate from the system packages. The separation a container provides go beyond just Python packages.
- While you can install another operating system on a second partition, this is not a container. Containers run on top of the operating system, but provide an independent environment for programs to run.

Check It!

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